Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 5

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

Team Members

Aditi Kapadiya	AU2440139
Disha Gadhiya	AU2440217
Khushi Panchal	AU2440002
Harshita Dhanwani	AU2440083

1. Overview and Updated Project Direction

In Week 5, the project direction was shifted from implementation and comparison of texture descriptors at a fixed scale to study multi-scale and adaptive classical approaches to fabric-defect localization. The new focus is to see if the information from the repeating pattern of woven fabrics is better suited for scale selection than the same information for each patch of fabric. This change is not intended to replace but is an extension of the earlier LBP, GLCM and Gabor baseline work.

The new Week 5 plan includes four primary changes: (1) representation of texture at multiple scales, (2) estimation of the major periodicity of the texture by autocorrelation and Fourier analysis, (3) combination of the scale-wise deviation maps, and (4) comparison of the fixed-scale and adaptive-scale pipelines using the same image-mask pairs. The methods are still classical and not based on a learned deep neural network.

2. Work Done in Week 5

Project methodology re-written: Workflow has been modified with more focus on fixed-versus-adaptive multi-scale comparison. The previous descriptors still serve as baseline descriptors, and periodicity scale selection is under investigation as the adaptive component.

Multi-scale Gabor pipeline developed: A gabor filter bank is applied to a number of different wavelengths and orientations. The filter responses are also turned into local deviation maps, normalized and combined into a candidate defect-response map.

The following two methods are used to estimate a dominant repetition period from a fabric patch: autocorrelation and one-dimensional Fourier power spectrum. The estimate is used to inform the selection of adaptive wavelengths. If the estimate is not reliable or is not possible, the pipeline reverts to the pre-defined fixed scales.

Multi-scale extensions of LBP and GLCM: a multi-radius deviation map of LBP is prepared and an optional multi-window GLCM map. Since local features calculated in dense manner are more expensive to compute, it is possible that they can be used in a useful manner, hence GLCM is treated as an optional experiment until it's usefulness and runtime is assessed.

Scale-wise response fusion prepared: Mean and maximum fusion are supported. A preliminary evaluation workflow consists of computing precision, recall, F1-score and intersection over union (IoU) at the pixel level with respect to binary masks.

The dataset validation and visualization explored: The current patch dataset was scanned for pairing of image-masks, dimensions of the images, variation of image and mask intensity levels, and the content of the masks. This showed that the first patch automatically selected was constant, and was therefore not suitable for meaningful periodicity estimation. A revision of the sample was thus made, favouring patches where there was visible texture.

3. Literature Review

Previous studies have shown that frequency domain and multi-scale texture analysis techniques are suitable for fabric inspection. The literature does not, however, suggest that a single method will be most successful on all fabrics and/or flaws. The planned comparison is thus an experiment.

3.1 Fourier analysis and periodic fabric structure

Chan and Pang were interested in fabric-defect detection, which is related to the regular and repetitive structure of fault-free fabric, and applied Fourier analysis to help them. Their work shows how defects can affect the spatial frequency of a cloth image. This assists in exploring the potential of investigating the autocorrelation and the use of periodicity estimates based on FFT to select scales that are appropriate for the fabric structure. One of the main reasons for choosing periodicity in the current project is for use in guiding the scales of local filters, which is a similar, but not identical, approach to the spectral analysis method in that paper.

For the reference, please read C.-H. Chan and G. K. H. Pang, “Fabric Defect Detection by Fourier Analysis,” IEEE Transactions on Industry Applications, vol. 36, no. 5, pp. 1267–1276, 2000. DOI: 10.1109/28.871274.

3.2 Gabor filters and multi-scale response fusion

The purpose of gabor filters is to capture the texture information at various spatial frequencies and different orientations. The authors Mak and Peng suggested a scheme to inspect textiles using Gabor filters tuned to non-defective fabric textures. In their later work, they use multi-level tuned-matched Gabor filters, which use the responses of multiple levels and take the trade-off between defect segmentation quality and computational cost into account. These studies serve as motivation for employing a filter bank, multi-scale response maps, and fusion, as well as pointing out that the more scales there are the more computation there will be, but not necessarily the more detection power.

The references include K. L. Mak and P. Peng, “An automated inspection system for textile fabrics based on Gabor filters”, Robotics and Computer-Integrated Manufacturing, 2008. DOI: 10.1016/j.rcim.2007.02.019. K.-L Mak, P Peng and K-F C Yiu, “Fabric defect detection using multi-level tuned-matched gabor filters”, Journal of Industrial and Management Optimization, vol 8, no 2, pp. 325–341, 2012. DOI: 10.3934/jimo.2012.8.325.

3.3 GLCM and statistical texture descriptors

Gray-level co-occurrence matrix (GLCM) is used to describe the spatial relationship between a pair of gray levels; it gives texture measures like contrast and homogeneity. A published comparison study between GLCM and Gabor methods for fabric-defect detection shows that both the statistical based methods and the filter based methods are reliable techniques available and the performance of them is dependent on the implementation and data. In this project, multi-window GLCM is used as a complimentary descriptor instead of being believed to outperform Gabor.

For the reference: “Fabric defect detection based on GLCM and Gabor filter: A comparison”, vol:124, Issue:23, Page:6467-6472, Optik. 6469–6474, 2013. DOI: 10.1016/j.ijleo.2013.05.004.

3.4 LBP as a local texture descriptor

Local Binary Patterns (LBPs) are based on local binary intensity comparisons and they are employed to describe micro-texture extensively. Research on fabric defects has included other texture descriptors besides LBP that are used for classification, and gabor responses are commonly employed for the detection and/or localization phase. With multiple LBP radii, it is possible to explore texture at a variety of neighborhood sizes. Within the current project, the LBP is used as a baseline/extension to compare with the periodicity-guided Gabor approach.

Fabric defect detection using gabor filters and defect classification based on lbp and tamura method, The Journal of The Textile Institute, vol. 104, no. 1, pp. 18–27. DOI: 10.1080/00405000.2012.692940.

4. Challenges Faced

Constant valued patches: The first image patch that was automatically selected, had all of its pixel values equal to 255. There is no period that could be returned from autocorrelation or FFT in a constant patch. The patches rejected by the sample-selection step must not be constant or almost constant.

Some patches cut from bad source images: If a patch is cut from a bad source image, there may be no defect region in the patch. They can have a mask to contain no defect pixels correctly. The patches need not automatically be seen as pairing failures, but the results need to be split into two categories: failure or no failure.

Mask interpretation and alignment: A mask can have defect pixels and not have any correct file name, or it can be misaligned and have a correct file name. The pipeline should validate dimension, validate mask values, and visually validate overlays prior to using metrics.

Periodicity estimates: The periodicity in the texture of the fabric may be hard to estimate if texture contrast is low, if the texture variation is from illumination changes or if there aren't enough repeated units in the patch. Therefore, the implementation requires a safe harbor to scales and a record of when there is a safe harbor to scales.

Computational cost: The computation of dense local GLCMs is costly especially when performed at multiple window sizes. For the actual method, a smaller set or a less detailed sampling grid must be used before it is applied to the whole patch collection.

The output is a continuous deviation map that needs to be thresholded to get a binary prediction; this is done with fair evaluation and threshold sensitivity. Precision and recall can vary depending on the threshold and fusion rules used. Fixed and adaptive pipelines should use the same evaluation set and similar thresholding rules; threshold parameters should not be optimized on the evaluation set, but on the validation set.

5. Present conditions and constraints

Notebook and experimental workflow for Week 5 is ready, which covers: periodicity estimation; multi-scale Gabor response maps; LBP/GLCM extensions; fusion options and metric calculations. Initial debugging has shown that before any output can be interpreted, patch selection and mask inspection must be done. The notebook is syntaxchecked and a complete validated run over the dataset has yet to be confirmed. For this reason, this report does not lead to a numerical accuracy or improvement of performance nor to a complete fixed versus adaptive result.

6. Plan of Work for Week 6

- Stabilize the periodicity estimator and evaluate its results with representative patches of fabrics of various constructions.
- Determine and validate fixed multi-scale and periodicity-adaptive configurations with a consistent validation set.
- Explore thresholding methods and include post-processing like morphological opening/closing and connected component filtering.
- Compute a score of defect confidence/reliability using deviation-map strength and/or region statistics.
- Examine failure cases, particularly low contrast defects, small defects, irregular texture or patches with no marked defects.
- Save representative input, ground-truth mask visualizations and deviation-map and predicted-mask visualizations for documentation.
- Continue the complete metric evaluation and fabric-wise/defect-wise evaluation as the next step in the evaluation, following the project plan.

References

1. C.-H. Chan and G. K. H. Pang, “Fabric Defect Detection by Fourier Analysis,” IEEE Transactions on Industry Applications, vol. 36, no. 5, pp. 1267–1276, 2000. https://doi.org/10.1109/28.871274
2. K. L. Mak and P. Peng, “An automated inspection system for textile fabrics based on Gabor filters,” Robotics and Computer-Integrated Manufacturing, 2008. https://doi.org/10.1016/j.rcim.2007.02.019
3. K.-L. Mak, P. Peng and K.-F. C. Yiu, “Fabric defect detection using multi-level tuned-matched Gabor filters,” Journal of Industrial and Management Optimization, vol. 8, no. 2, pp. 325–341, 2012. https://doi.org/10.3934/jimo.2012.8.325
4. “Fabric defect detection based on GLCM and Gabor filter: A comparison,” Optik, vol. 124, no. 23, pp. 6469–6474, 2013. https://doi.org/10.1016/j.ijleo.2013.05.004
5. “Fabric defect detection using Gabor filters and defect classification based on LBP and Tamura method,” The Journal of The Textile Institute, vol. 104, no. 1, pp. 18–27. https://doi.org/10.1080/00405000.2012.692940